Tannins

- General Properties
- Classification of Tannins
  - Hydrolysable Tannins
  - Nonhydrolysable Tannins(condensed)
  - Pseudotannins
- Chemistry of Tannins

Introduction

- Plants accumulate a wide variety of "secondary" compounds, including alkaloids, terpenes and phenolics.
- Although these compounds apparently do not function in "primary" metabolism such as biosynthesis, biodegradation and other energy conversions of intermediary metabolism, they do have diverse biological activities ranging from toxicity to hormonal mimicry, and may play a role in protecting plants from herbivory and disease.
- Phenolic metabolism in plants is complex, and yields a wide array of compounds ranging from the familiar flower pigments (anthocyanidins) to the complex phenolics of the plant cell wall (lignin).

- However, the group of phenolic compounds known as tannins is clearly distinguished from other plant secondary phenolics in their chemical reactivities & biological activities
- Tradition use of tannins as agents for converting animal hides to leather ("tanning") is one manifestation of the most obvious activity of the tannins: their ability to interact with and precipitate proteins, including the proteins found in animal skin.

- The term "tannin" comes from the ancient Celtic word for oak, a typical source for tannins for leather making.
- Tannins are one of the most widely occuring group of natural substances in different families of higher plants.
- These secondary metabolites are present in solution form in the cell sap and also in distinct vacuoles.
- They have been known since long time as the astringent substances, having the capacity to combine with tissue proteins and precipitate them.

TANNINS

- Historically, the importance of tannin-containing drugs is linked to their tanning properties (ability to transform fresh hides into leather).
- These included chestnut tree tannin (*Castanea sativa* L), oak tannin (*Quercus robar* L) in Europe, but also, in other parts of the world, tannins from Anacardiaceae (Quebracho – *Schinopsis spp.*), sumacs (*Rhus spp.*), Legumes (acacias – *Acacia spp.*) or Combretaceae (myrobalans – *Terminalia spp.*).

Definition

- A more recent definition of tannins: "Phenolic natural products that precipitate proteins from their aqueous solutions".

- "water-soluble phenolics of molecular weight between 500 and 3000, which, in addition to displaying the classic reactions of phenols, can precipitate alkaloids, gelatine, and other proteins."

General Properties

- They are pale yellow to light brown-red amorphous substances widely distributed in plants and used chiefly in tanning leather, dyeing fabric, and making ink.
- Tannins are found as shapeless yellowish or light brown masses like powder, flakes or sponge.
- Their solutions are acid and have an astringent taste.
- They are isolated from oak bark, sumac, and galls.
- Tannins give tea astringency, colour, and flavour.
- Tannins are used industrially to clarify wine and beer, reduce viscosity of oil-well drilling mud, and they have also had medical uses.

General Properties

- Tannins are mainly found in the cell vacuoles or in the cuticle (here cells don't interfere with plant processes: only after death and cell breakdown do they become active in metabolic processes).
- Tannins are phenolic glycosides.
- Water-soluble
- Also soluble in dilute alkali, alcohol, glycerol and acetone
- Slightly soluble in other organic solvents.
- Solutions precipitate heavy metals, alkaloids, glycosides and gelatin.

Identification of tannins

1-Tannin + $FeCl_3$ → bluish black colour with hydrolysable tannin and brownish black colour with catechol.

2-**Gelatin test**: solution of 1% tannin precipitate gelatin from its solution at conc. 1%.

3-**Gold beater's skin test**: soak a small piece of Gold beater's skin membrane in 2% HCl, rinse with distilled H_2O, place the piece for 5 minutes in the tannin solution to be tested, wash with H_2O, transfer to 1% $FeSO_4$ solution, a brown colour is formed.

4-**Bromine water**: only condensed tannin gives ppt. with bromine water.

Chemical Properties of Hydrolysable & Condensed

- With ferric salts, gallo-tannins and ellagi-tannins give blue-black precipitates.
- Condensed tannins give brown-green precipitates.
- Condensed tannins turn red with vanillin and HCL.
- Gallo-tannins give a pink colour with potassium iodate.
- Ellagitannins react (are coloured) with nitrous acid in the presence of acetic acid. Colour reaction: pink → purple → blue.
- TLC is used to analyze extracts (visualized under UV light and with the reagents mentioned above)
- HPLC can also be used.

Tanning Properties

- Formation of bonds between the collagen fibres of the hide (imparts resistance to water, heat and abrasion.)
- This capability of tannins to combine with macromolecules explains why they precipitate cellulose, pectins and proteins; it also explains their characteristic astringency and tartness by precipitating the glycoproteins contained in saliva, tannins make the latter lose its lubricating power.

- The combination between tannins and macromolecules is established by hydrophobic interactions and hydrogen bonds between the phenolic groups of tannins and the proteins or other polymers.
- Other types of linkages (irreversible) must also be involved to ensure the lasting stability of the combination between tannins and collagen structures.
- These include the covalent bonds established after oxidation of the phenols to quinones.
- The condition necessary for the formation of these linkages is the tannin's molecular weight must fall within a well defined range. If it is too high, the molecule cannot insert itself into the interfibrillar spaces of the macromolecule. If it is too low, the molecule can insert itself but cannot form enough bonds to stabilize the combination.

Tannin Incompatibilities

- Clove and cinnamon contain tannins and also other principal therapeutic constituents.
- This may complicate extraction or produce imcompatibilities with other drugs.

  E.g. Many alkaloids for example are precipitated by tannins.

Tannin Extraction

- Tannins are extracted using water and acetone.
- Optimal yields are obtained from fresh tissues or freeze-dried tissues.
- Optimal yields are not obtained from dried tissues (tannins are irreversibly combined with other polymers)

- After eliminating the acetone (distillation), the pigments and lipids are removed from the aqueous solution by a solvent extraction.

- Ethyl acetate extraction of the aqueous solution separates the dimeric proanthrocyanidins and most gallotannins.

- Polymeric proanthrocyanidins and high molecular weight gallotannins remain in the aqueous phase.

- To obtain pure compounds, the appropriate chromatographic techniques are used (most often one of the gel filtrations)

- Followed by reverse-phase chromatography (in water and alcohol, or water, alcohol and acetone mixures)

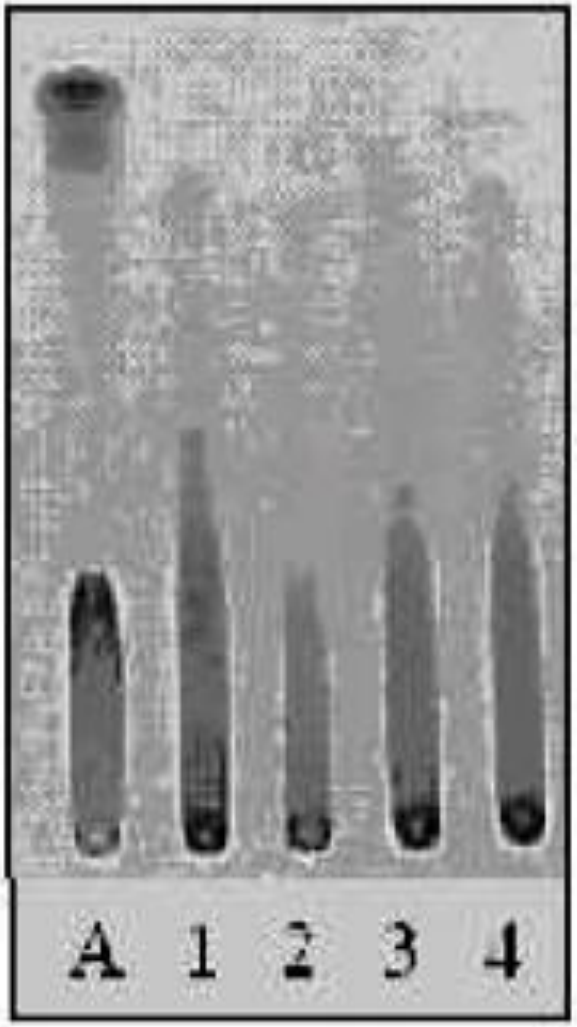


Figure 1 - TLC ethanol extract from acacia bark (1-dry and milled; 2-dry and entire; 3-natural and milled; 4-natural and entire; A-tannic acid standard)

Medicinal Properties

- Therapeutic/medicinal activities result due to the astringency properties.
- Tannin containing herbs (drugs) will precipitate proteins.
- Externally: waterproof the external layers of the skin and mucosa: Protect underlying layers.
- Vasoconstrictor effects – on small superficial vessels
- Limits fluid loss – enhance tissue regeneration in superficial wounds and burns
- Internally: Anti-diarrhoeal
- Anti-septic effects (anti-bacterial/anti-fungal) → used to Rx infectious diarrhoeas and dermatitis.

Anti-oxidant Activity of Tannins

- Hydrolysable tannins (more so than condensed tannins) are free-radical scavengers.
- Inhibit superoxide ion formation.
- Some inhibit lipoxygenase
- They do not inhibit cyclo-oxygenase.
- Anti-oxidant flavonols and proanthrocyanidins from grape juice and wine are widely considered to be the main principles responsible for the preventative effect of CVD.

Chemistry of hydrolysable tannins

- Members of this class consist of several molecules of phenolic acids united by ester linkage to a central glucose residue. Such esters are readily hydrolyzed by acids or enzymes so called hydrolysable tannins.
- Types of hydrolysable tannin:

A- Gallitannin

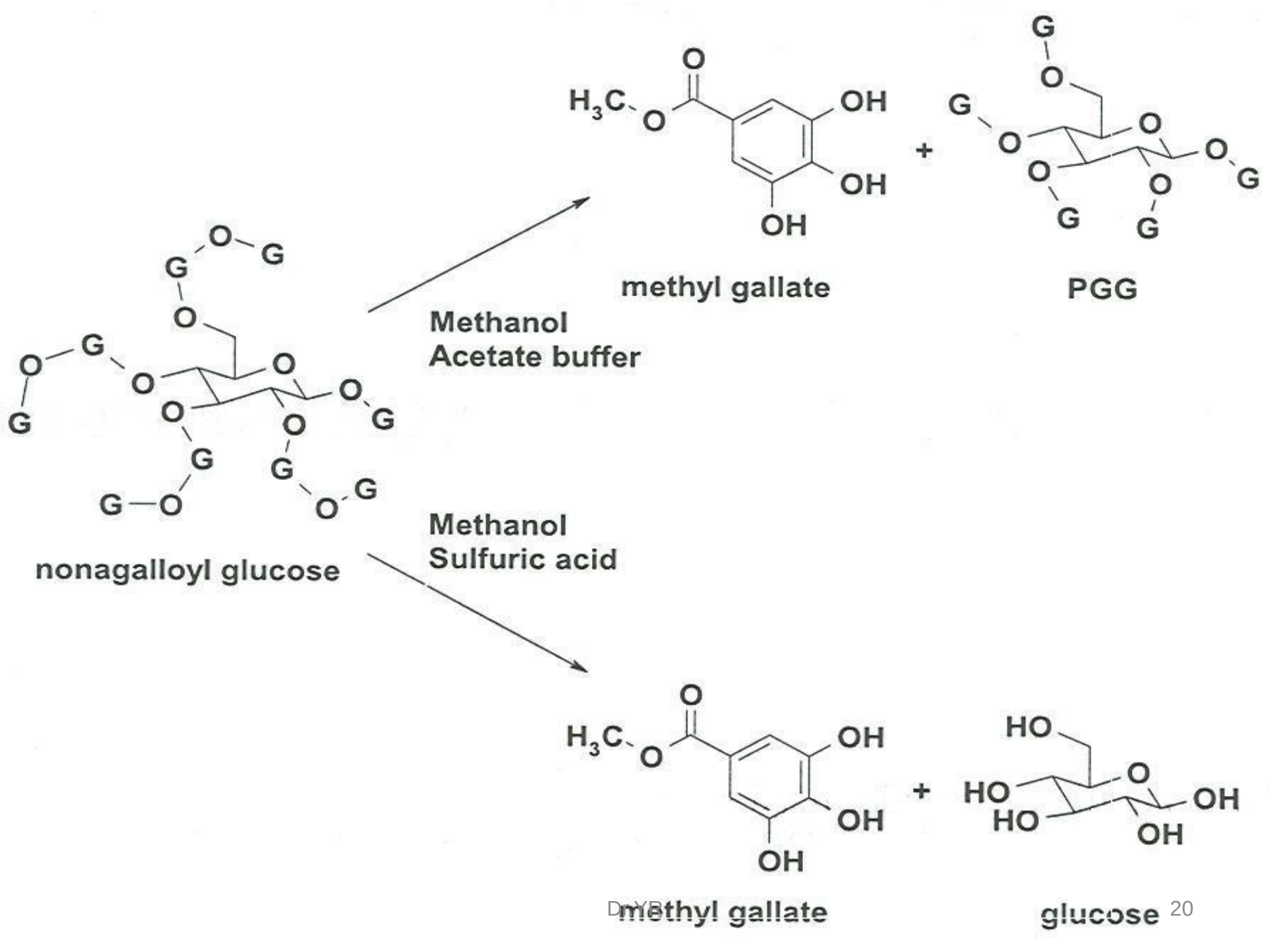
On hydrolysis , it gives gallic acid and glucose.

Present in rhubarb, galls, clove, hamamelis.

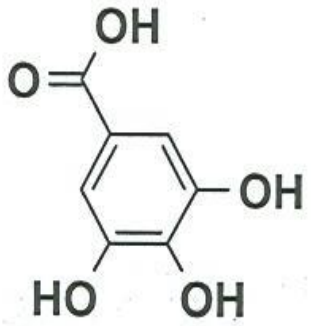
B- Ellagitannin

On hydrolysis it gives ellagic acid and glucose

Present in pomegranate and eucalyptus.

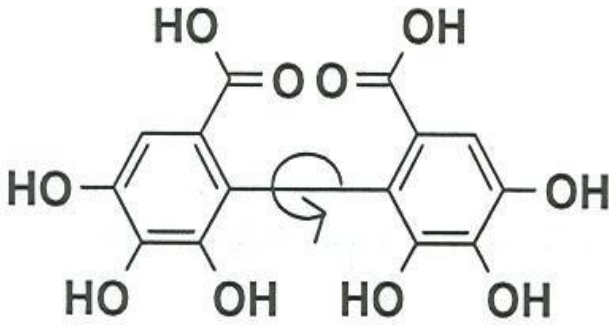

nonagalloyl glucose
Methanol
Acetate buffer
methyl gallate
PGG
Methanol
Sulfuric acid
methyl gallate
glucose

gallotannin →

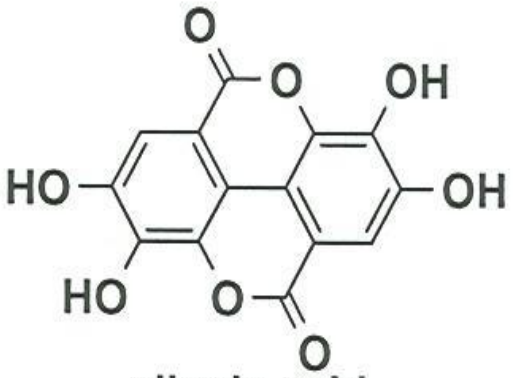


gallic acid

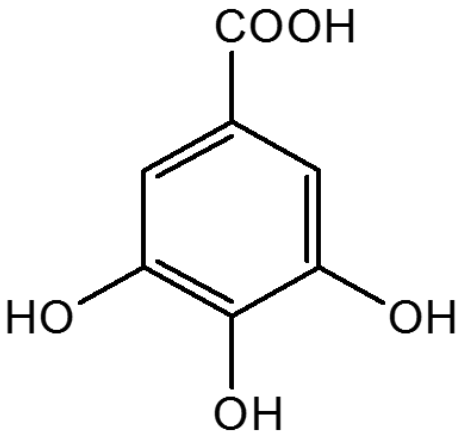
ellagitannin →



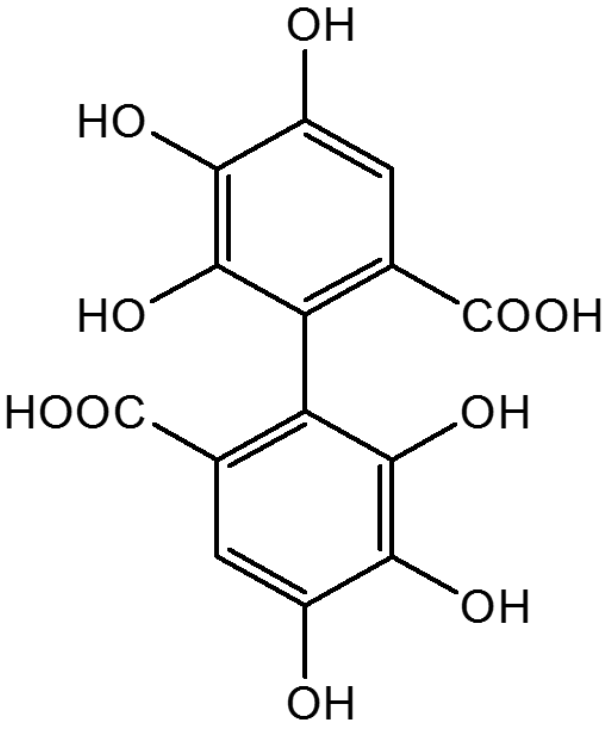
hexahydroxydiphenic acid
(HHDP)



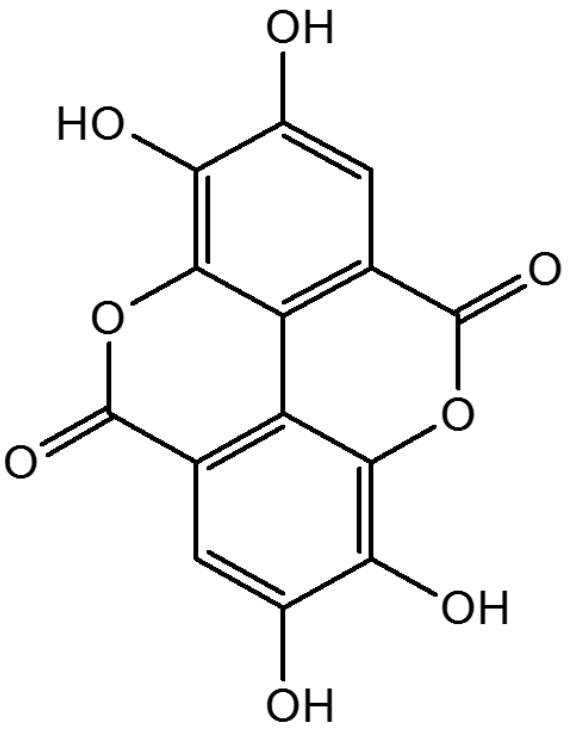
ellagic acid



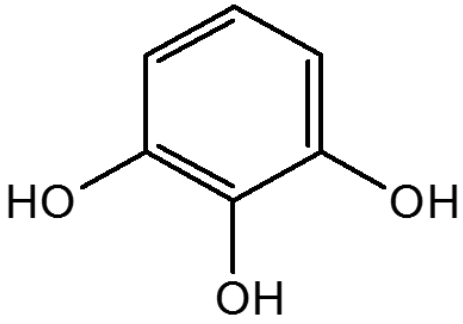
Gallic acid



Hexahydroxydiphenic acid



Ellagic Acid



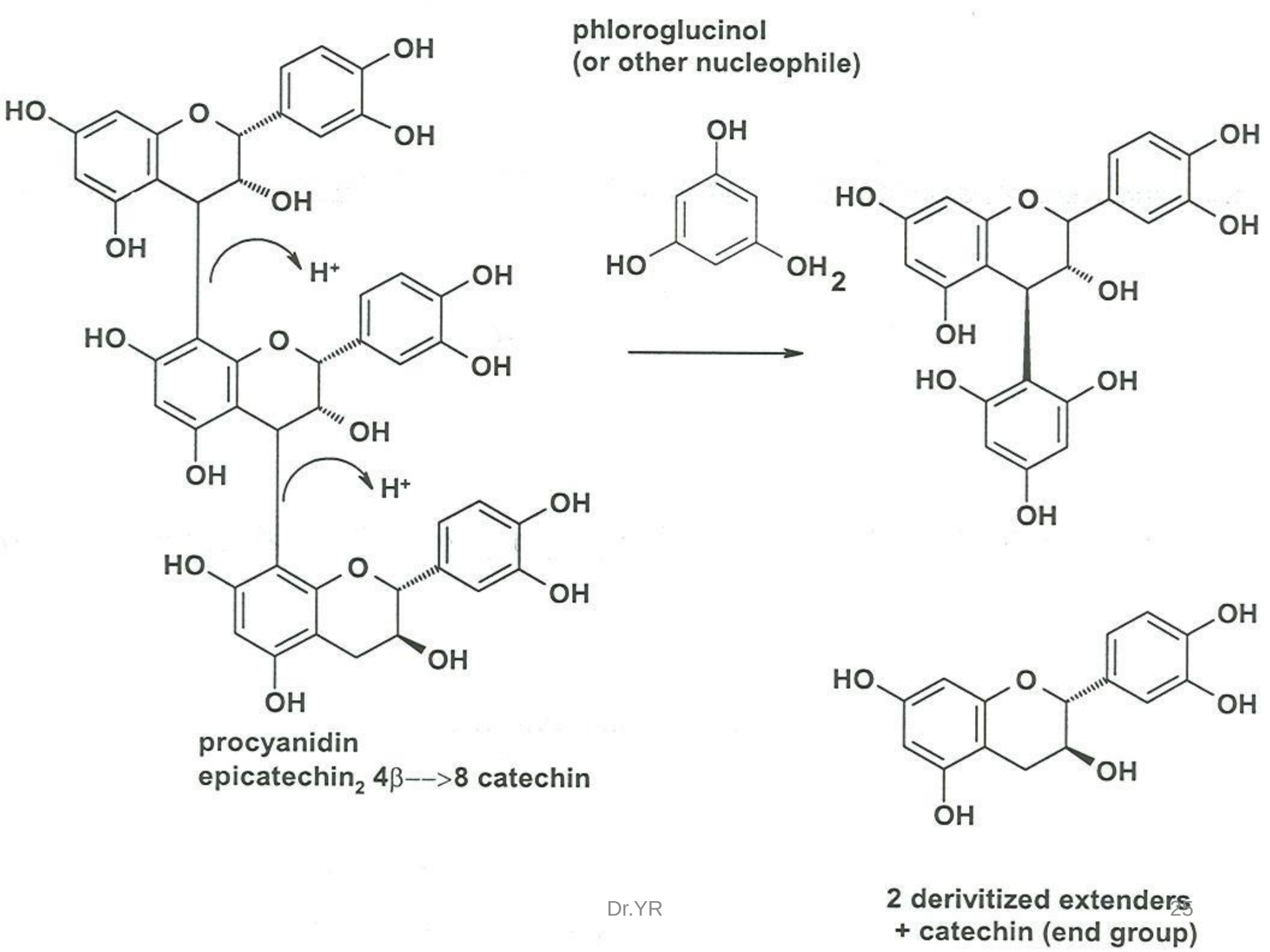
Pyrogallol

Chemistry of condensed tannins

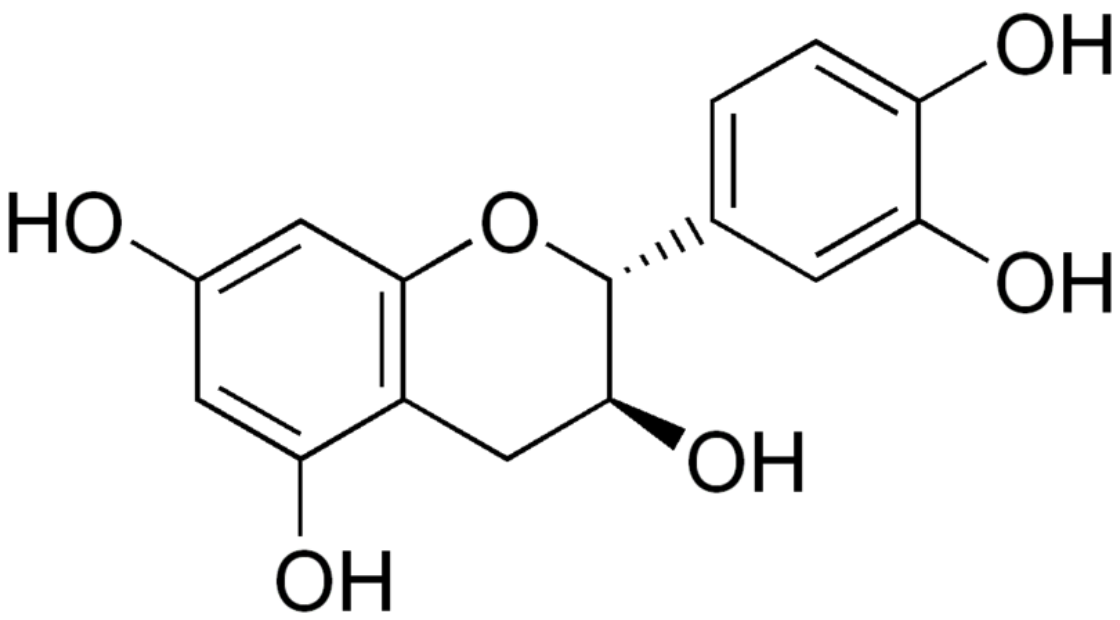
- Proanthocyanidins (condensed tannins) are polymeric flavanoids. The flavanoids are a diverse group of metabolites based on a heterocyclic ring system derived from phenylalanine (B) and polyketide biosynthesis (A).
- Although the biosynthetic pathways for flavanoid synthesis are well understood, the steps leading to condensation and polymerization have not been elucidated.
- The flavanoid skeleton, the standard letters to identify the rings, and the numbering system are shown here.

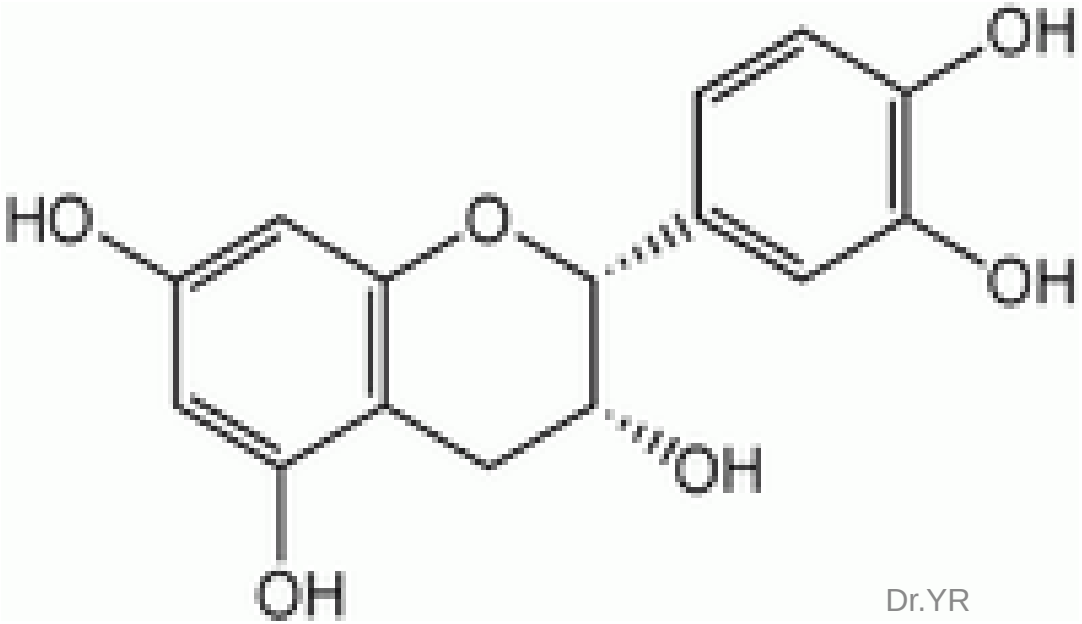
Condensed tannin

- ❖On dry distillation they give catechol.
- ❖They differ from pyrogallol tannin where their structures are related to flavonoids and they lack sugar in their structures
- ❖Present in cinnamon, cinchona, tea.
- ❖Catechin:Flavan-3-ol and leucocyanidin: flavan-3,4-diol.
- The best characterized condensed tannins are linked via a carbon-carbon bond between C_8 of the terminal unit and C_4 of the extender.


phloroglucinol
(or other nucleophile)
procyanidin
epicatechin2 4β-->8 catechin
2 derivitized extenders
+ catechin (end group)

Catechin





epicatechin

Pseudotannins

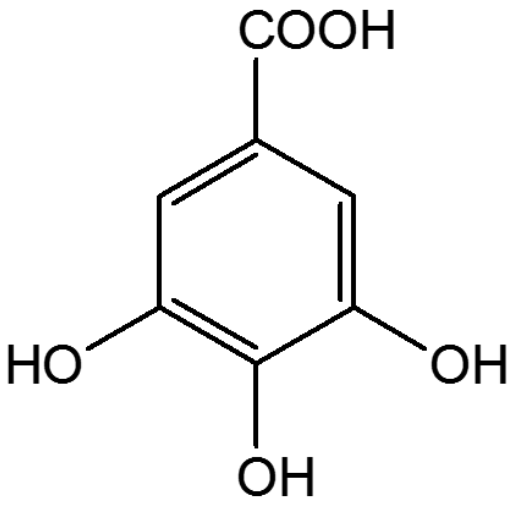
- They are compounds of lower molecular weight than true tannins and they do not respond to the *goldbeater's skin test*.
- Examples of drugs containing Pseudotannins are:

  Gallic acid: Rhubarb

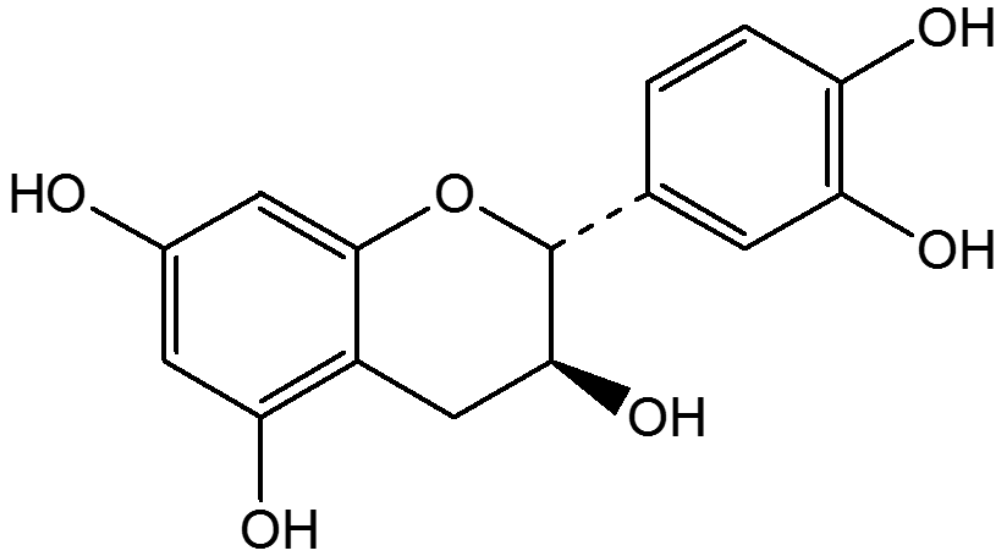
  Catechins: Guarana, Cocoa

  Chlorogenic acid: Mate, Coffee

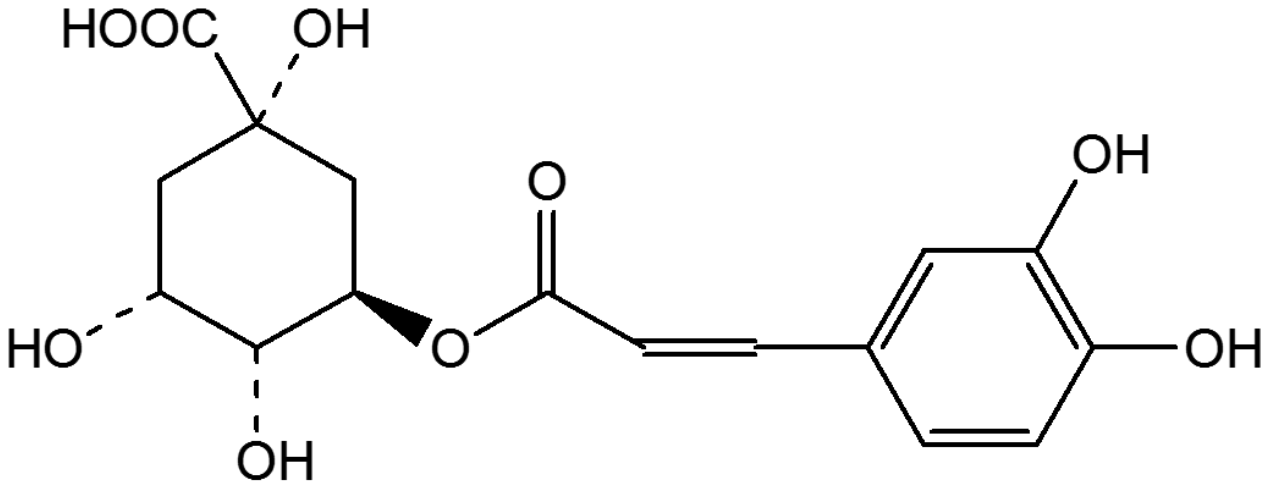
  Ipecacuanhic acid: ipecacuanha



Gallic acid



Catechin



Chlorogenic acid

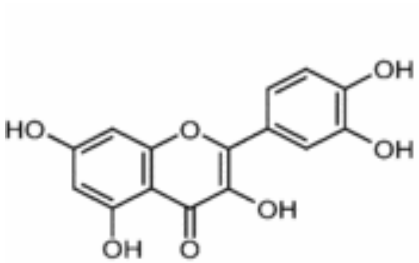
True Tannins Vs Pseudo-Tannins

True tannins:

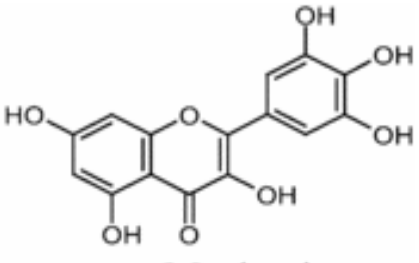
Molecular weights of 1000-5000.

Pseudo-tannins:

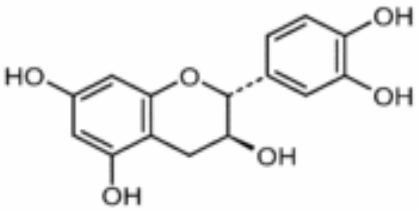
Gallic acid, catechins and chlorogenic acid are similar phenolic compounds: have a low molecular weight.



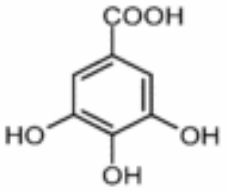
Quercetin



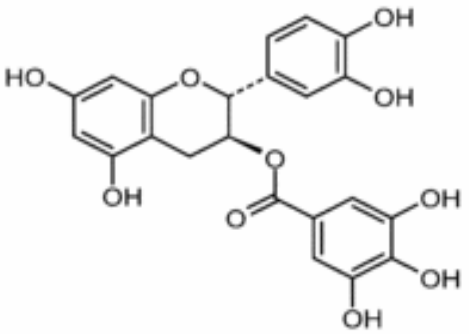
Myricetin



Catechin



Gallic acid



Catechin-gallate

Classification of Tannins

- In higher plants, two groups of tannins are generally distinguished, which differ by their structure, as well as their biogenic origin – hydrolysable tannins and condensed tannins.

Hydrolysable Tannins: Ellagitannins

Gallotannins

Condensed tannins: Catechin

Hydrolysable Tannins

- These tannins on heating with hydrochloric or sulphuric acids yield gallic or ellagic acids.
- They may be **hydrolyzed by acids** or **enzymes** such as **tannase**, (products being sugar and phenolic acids).
- Formed by several phenolic acids which are united by ester linkages to a central sugar (glucose) molecule.
- Hydrolysable tannins are therefore **esters of a sugar and phenolic acid molecules**.
- The phenolic acid is
  - in the case of **gallotannins – gallic acid**
  - in the case of **ellagi-tannins** – hexahydroxydiphenic acid (HHDP) and its **oxidized derivatives** (dehydrohexahydroxydiphenic acid).
  - Ellagitannins are found in Herb Robert (Geranium species, Meadowsweet and Oak bark).

Hydrolysable Tannins

- Hydrolyzable tannins are hydrolyzed by weak acids or weak bases to produce carbohydrate and phenolic acids.
- Examples of gallotannins are the gallic acid esters of glucose in tannic acid, (found in leaves and bark of many plant species)

Gallotannins: rhubarb, cloves, red rose petals, bearberry leaves, Chinese galls, Turkish galls, hamamelis, chestnut and maple

Ellagitannins: pomegranate rind, pomegranate bark, eucalyptus leaves, some Australian kinos, chestnut (Castanea spp), and oak bark.

- On dry distillation gallic acid and other components get converted to pyrogallol.
- They respond to ferric chloride solution producing blue colour.
- The examples of hydrolysable tannins are gallotannin in nutgall, rhubarb, clove and chestnut; ellagitannin from oak, Myrobalan, and pomegranate.

Gallotannins

- Gallotannins are the simplest hydrolysable tannins, containing a polyphenolic and a polyol residue.
- Polyol: alcohol with multiple hydroxyl groups.
- Most of the gallotannins isolated from plants contain a polyol residue derived from D-glucose.

Ellagitannins

- With more than 500 natural products characterized so far, ellagitannins are the largest group of known tannins.

- **E**llagitannins are formed from the gallotannins by the oxidative coupling of at least two galloyl units.

Plants rich in Hydrolysable Tannins

S.No.	Part of plant	Name of the plant
1	Barks	Arjuna bark (Terminalia arjuna)
2	Seeds	Oak galls (Quercus infectoria)
3	Fruits	Myrobalan(Terminalia chebula), Bahera(Terminalia belerica), Amla (Emblica officinalis) (indian goose berry)

Arjuna bark

- **Biological source**: It consists of dried stem bark of the plant known as *Terminalia arjuna* Rob.
- **Family**: Combretaceae.
- **Geographical source:** The tree is common in indian peninsula. It is grown by the side of streams and very common in central region of india.
- **Morphological characters:**

Colour	-	Greyish-brown
Odour	-	None
Taste	-	Astringent
Shape	-	Flat
Size	-	15 cm long, 10 cm wide and 1cm thick.

- **Chemical constituents**:
  - Arjuna bark contain 15% of tannins.
  - It also contains triterpenoid saponin (Arjunolic acid).
  - In addition it contains ellagic acid and arjunic acid.
- **Uses**:
  - Used as cardiotonic.
  - It is also febrifugel and antidysenteric.
  - It possesses diuretic(arjunolic acid) and tonic properties.
  - The drug exhibits hypotensive action with vasodialation and decreased heart rate.
  - This bark is used extensively in the tanneries and also as a timber.

Myrobalan

- **Biological source**: It consists of dried ripe and fully matured fruits of *Terminalia chebula*
- **Family**: Combretaceae.
- **Morphological characters:**

Colour	-	Yellowish brown
Odour	-	Odourless
Taste	-	Astringent, slightly bitter and sweetish at the end.
Shape	-	Ovate and wrinkled longitudinally.
Size	-	20 to 25mm long and 10 to 25mm wide.

- **Geographical origin:** It is found in forests of india.
- **Chemical constituents**:
  - Myrobalan fruits contain 25-32% of tannin.
  - They are pyrogallol, which on hydrolysis yield chebulic acid and d-galloyl glucose.
  - Chebulagic, chebulinic, ellagic and gallic acids are also present.
  - It also contains glucose and sorbitol.
- **Uses**: Used as astringent, laxative, stomachic and tonic.

Amla fruit

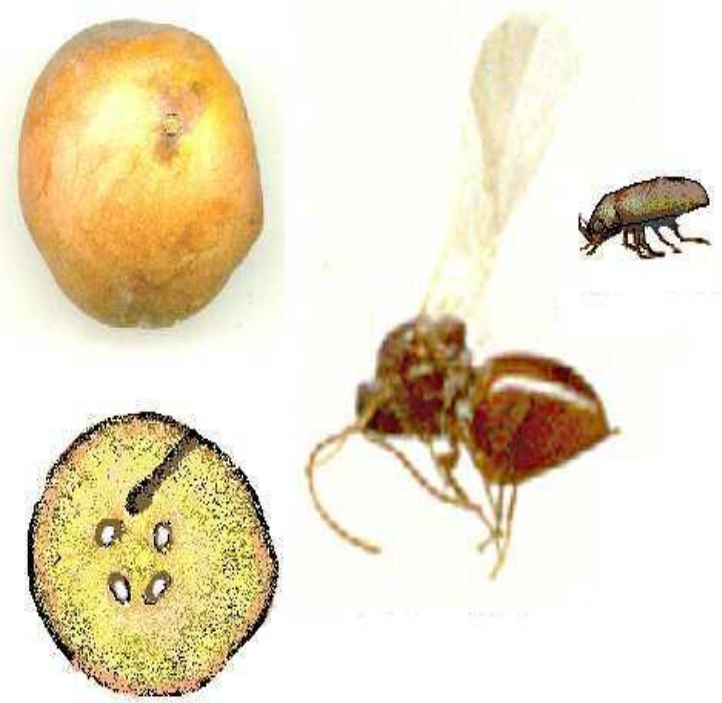
- **Biological source**: It consists of dried as well as fresh fruits of the plant *Emblica officinalis* Gaerth
- **Family**: Euphorbiaceae.
- **Geographical source:** This tree is found in india, srilanka and myanmar.
- **Morphological characters:**

Colour	-	Green changing to light yellow or brick red when matured
Odour	-	None
Taste	-	Sore and Astringent
Shape	-	the fruits are depressed, glabose.
Size	-	1.5 to 2.5cm in diameter.

- **Chemical constituents**:
  - Amla is rich source of vitamin C and contains 600-750mg per hundred g of the fresh pulp.
  - Apart from that it contain 5% tannin.
  - Tannins retards oxidation of vitamin C.
- **Uses**:
  - Used as diuretic and laxative.
  - They are also used in jaundice, anaemia along with iron compounds.
  - Fruits are used in preparation of inks, hair oils and shampoos.

Turkish Galls

Definition: Turkish galls (Turkey galls, Galla): Are vegetable growths formed on the young twigs of the dyer's oak, *Quercus infectoria* (Fagaceae). They result from the decomposition of the eggs of the gall-wasp *Adleria gallaetinctoriae*.



If the galls are not collected and dried, the mature insect bores out of the gall and escapes.

History: Ancient writers recorded the use of an infusion to test for sulphate of iron, possibly the earliest mention of an attempt to detect adulteration by chemical means.

Tannin containing drugs

Galls (Blue Galls, Aleppo Galls, Turkish Galls)

Origin: dried excrescence (pathological outgrowth) resulting from the deposition of the eggs of *Cynips gallae tinctoria* F. Cynipidae on the young twigs of *Quercus infectoria* F. Fagaceae. Collected before the escape of the insect and known as blue galls.



Morphology: subspherical, 12-20 mm in diameter, external surface is bluish green to olive green in colour, showing blunt short projections. Internally it is buff in colour.

Active constituents: 50-70% pyrogallol tannin: gallotannic acid, gallic acid and ellagic acid.

Uses: **Medicinally**: astringent, haemostatic used in treatment of hemorrhoids in form of suppositories or ointments.

Industrially: 1- in manufacture of ink and paints

2- in tanning of leather.

Chemical test: $FeCl_3$ → bluish black colour.

Turkish Galls: Collection

Galls are collected by peasants of Turkey and Syria.

After collection they are graded according to colour into 3 grades: blue, green and white. These are then found on the London Market.

Colour changes occur due to enzymatic secretions of insect.

- Uses
  - Used as a source of tannic acid.
  - Tannic acid is used as an astringent and styptic.
  - Used for tanning and dyeing.
  - Ink manufacturing.

Condensed Tannins

- They are also called as non-hydrolysable tannins, phlobatannins or proanthocyanidins.
- They are much resistant to hydrolysis.
- They are related to flavonoid pigments, because they are formed via derivatives of flavones, like catechin or flavon-3-ol or flavan-3,4-diols.
- Unlike the hydolysable tannins, on treatment with enzymes or mineral acids, they are polymerised or decomposed into red coloured substances called phlobaphenes, which are insoluble in water and indicate the typical brownish-red colour of many plants and drugs.

- Are polymers of 2 to 50 (or more) flavonoid units that are joined by carbon-carbon bonds, which are not susceptible to being cleaved by hydrolysis.
- All hydrolyzable tannins and most condensed tannins are water soluble - some very large condensed tannins are insoluble.
- On Dry distillation they yield catechol.
- Tannins with ferric chloride produces brownish-green colour.
- They are distributed in different parts of plants.
- The green tea and hamamelis leaves; cinchona, cinnamon and wild cherry bark; male fern rhizome; coca, cola and areca seeds; pale and black catechu contain these types of tannins.

Condensed Tannins

- Condensed tannins are oligomeric and polymeric proanthocyanidins consisting of coupled flavan-3-ol (catechin) units.
- (oligomeric or polymeric proanthocyanidins =condensed proanthocyanidins = condensed tannins).
- Biosynthetically the condensed tannins are formed by the successive condensation of the single building blocks (up to more than 50 may be included in a single condensed tannin).
- The coupling pattern of the catechin units in condensed tannins can vary considerably.

NOTE:

Oligomeric: a polymer that consists of 2, 3, or 4 monomeres.

Polymer:Any of numerous natural and synthetic compounds of usually high molecular weight consisting of up to millions of repeated linked units, each a relatively light and simple molecule.

Condensed Tannins/ Proanthrocyanidins

- They are not readily hydrolysed to simpler molecules and they do not contain a sugar moiety (unlike hydrolysable tannins).
- On treatment with acids or enzymes, condensed tannins are converted into red insoluble compounds known as phlobaphenes.
- Depending upon the source of the tannins, various modified methods are used for extraction.
- Methyl alcohol, hot water, acetone and ethyl acetate are the common solvents used for extraction, the extract is filtered and dried under vacuum.

Condensed Tannins/ Proanthrocyanidins

- Phlobaphenes give the characteristic red colour to many drugs such as red cinchona bark, which contains these phlobatannins and their decomposition products.
- These tannins are therefore sometimes called catechol tannins (as they yield catechol).
- Like catechol itself, their (condensed tannin) solutions turn green with ferric chloride.

Plants rich in Condensed Tannins

S.No.	Part of plant	Name of the plant
1	Barks	Cinnamon, Wild cherry, Cinchona, Willow, Hamamelis, Acacia (wattle,mimosa), Oak bark
2	Flowers	Lime and Hawthorn
3	Seeds	Cocoa, Guarana, Kola
4	Fruits	Cranberries, Grapes, Hawthorn
5	Leaves	Hamamelis, Hawthorn, Tea leaves (especially green tea)
6	Extracts and dried juices	Catechu, Acacia, East Indian kino and Eucalyptus kino

Ashoka bark

- **Biological source**: It consists of dried stem bark of the plant *Saraca indica* Linn.
- **Family**: Leguminosae.
- **Geographical source:** It is distributed in South Asia, i.e. in Malaysia, Indonesia, Sri Lanka and India.
- **Morphological characters:**

Colour	-	Outer side dark brown or almost black
		Internally it is reddish brown in colour
Odour	-	None
Taste	-	Astringent and bitter.
Shape	-	longitudinal.
Size	-	variuos sizes upto 50cm length and 1cm in thickness.

- **Chemical constituents**:
  - Ashoka bark contains about 6% of tannins.
  - It also contains haematoxylin, ketosterol, saponin and organic calcium and iron compounds.
- **Uses**:
  - Used as uterinetonic and also as sedative.
  - It stimulates the uterus by prolonged and frequent uterine contractions.
  - It is also suggested in all cases of uterine bleeding.
  - It is reported to have a stimulant effect on the endometrium and ovarian tissue and useful in menorrhagia.

Pale Catechu

- Biological source: Gambir or pale catechu is a dried aqueous extract prepared from the leaves and young twigs of a climbing shrub, *Uncaria gambir*
- Family: Rubiaceae.
- Geographical source:

  The plant is Native to Malaysia singapore and Indonesia.
- Uses: Astringent, Diarrhoea.

  Mainly used in dyeing and tannin industries.





- **Morphological characters**

  Colour - Reddish-brown cuboidal mass

  Odour - None

  Taste - Astringent.

- **Chemical constituents**
- It contains condensed tannins, Catechins(7-33%), Catechutannic acid(22-50%) and Catechu red
- The drug also contains Quercetin and Gambir-florescin.

Pseudo-Tannins

- Pseudo-tannins are compounds of lower molecular weight than true tannins and they do not respond to the goldbeater's skin test.
- Gallic acid: rhubarb
- Catechins: catechu, acacia cutch, many Australian kinos, cocoa, guarana and many other drugs containing condensed tannins
- Ipecacuanhic acid: ipecacuanha

Pseudo-Tannins

- Chlorogenic acid: coffee (particularly unroasted) and Nux-vomica (a small quantity only).

  The detection test for chlorogenic acid is carried out by extracting the drug with water and treating this extract with ammonia solution, followed by exposure to air, which leads slowly to formation of green colour.

Tannin Function

- Tannins are of wide occurrence in plants and are usually found in greatest quantity in dead or dying cells. They exert an inhibitory effect on many enzymes due to protein precipitation and, hence, may contribute a protective function in barks and heartwoods.

